

Exercise_1

September 7, 2026

Give the output of the following expressions

(a) $1 + 3 * 2 - 5 + 4$

(b) $1 + 3 * (2 - 5) + 4$

(c) $5 ** 2 ** 2 * 3 + 1$

(d) $5 * (2 * 2) * 3 + 1$

(e) $1 + 3 / 2$

```
[1]: # (a)
      1 + 3 * 2 - 5 + 4
```

[1]: 6

```
[2]: # (b)
      1 + 3 * (2 - 5) + 4
```

[2]: -4

```
[3]: # (c)
      5 ** 2 ** 2 * 3 + 1
```

[3]: 1876

```
[4]: # (d)
      5 ** (2 ** 2) * 3 + 1
```

[4]: 1876

```
[5]: # (e)
      1 + 3 / 2
```

[5]: 2.5

Give the expressions & output of the following:

(a) 10^5

(b) $\sqrt{10}$

(c) the roots of $x^2 - 7x + 10 = 0$

(d) $\log(2 + \sqrt{5})$

(e) the area of a circle with a radius of 5.5

(f) $\sin 2.5$

(g) the complex roots of $x^2 - 2x + 10 = 0$

(h) $4!$

(i) $\sum_{k=32}^{128} k$

(j) $\prod_{k=3}^{17} k$

```
[6]: # (a)
     10 ** 5
```

```
[6]: 100000
```

```
[7]: # (b)
     10 ** 0.5
```

```
[7]: 3.1622776601683795
```

```
[8]: # (c)
import math
print("Using Quadratic Frmula \n")
D = 7 ** 2 - 4 * 1 * 10
root1 = (7 + math.sqrt(D)) / 2
root2 = (7 - math.sqrt(D)) / 2
print("Roots are : ",round(root1,2)," & ",round(root2,2))
```

Using Quadratic Frmula

Roots are : 5.0 & 2.0

```
[9]: # (d)
import math
math.log10(2 + math.sqrt(5))
```

```
[9]: 0.6269629207499362
```

```
[10]: # (e)
import math
radius = 5.5
math.pi * radius ** 2
```

```
[10]: 95.03317777109125
```

```
[11]: # (f)
import math
math.sin(2.5)
```

```
[11]: 0.5984721441039564
```

```
[12]: # (g)
complex_roots = [1 + 3j, 1 - 3j]
```

```
complex_roots
```

```
[12]: [(1+3j), (1-3j)]
```

```
[13]: # (h)
import math
math.factorial(4)
```

```
[13]: 24
```

```
[14]: # (i)
sum(range(32, 129))
```

```
[14]: 7760
```

```
[24]: # (j)
import math
math.prod(range(3, 18))
```

```
[24]: 177843714048000
```

Write a program that asks for user input and performs

- (a) Celsius to Fahrenheit conversion.
- (b) Fahrenheit to Celsius conversion.
- (c) Meter to foot conversion.
- (d) Foot to meter conversion.
- (e) Acre to square meter conversion.
- (f) Square meter to acre conversion.
- (g) Pound to kilogram conversion.
- (h) Kilogram to pound conversion.

```
[16]: # (a)
# Formula:  $F = (C * 9 / 5) + 32$ 
C = float(input("Enter temperature in Celsius: "))
F = (C * 9 / 5) + 32
print("Temperature in Fahrenheit =", F)
```

Temperature in Fahrenheit = 95.9

```
[17]: # (b)
# Formula:  $C = (F - 32) * 5 / 9$ 
F = float(input("Enter temperature in Fahrenheit: "))
C = (F - 32) * 5 / 9
print("Temperature in Celsius =", C)
```

Temperature in Celsius = 37.05555555555556

```
[18]: # (c)
# Formula:  $ft = m * 3.28084$ 
```

```
m = float(input("Enter length in meters: "))
ft = m * 3.28084
print("Length in feet =", ft)
```

Length in feet = 18.04462

```
[19]: # (d)
# Formula: m = ft * 0.3048
ft = float(input("Enter length in feet: "))
m = ft * 0.3048
print("Length in meters =", m)
```

Length in meters = 1.2192

```
[20]: # (e)
# Formula: m2 = ac * 4046.8564224
ac = float(input("Enter area in acres: "))
m2 = ac * 4046.8564224
print("Area in square meters =", m2)
```

Area in square meters = 18210.8539008

```
[21]: # (f)
# Formula: ac = m2 / 4046.8564224
m2 = float(input("Enter area in square meters: "))
ac = m2 / 4046.8564224
print("Area in acres =", ac)
```

Area in acres = 0.013590795980694093

```
[22]: # (g)
# Formula: kg = lb * 0.45359237
lb = float(input("Enter weight in pounds: "))
kg = lb * 0.45359237
print("Weight in kilograms =", kg)
```

Weight in kilograms = 5.4431084400000005

```
[23]: # (h)
# Formula: lb = kg / 0.45359237
kg = float(input("Enter weight in kilograms: "))
lb = kg / 0.45359237
print("Weight in pounds =", lb)
```

Weight in pounds = 2.2046226218487757